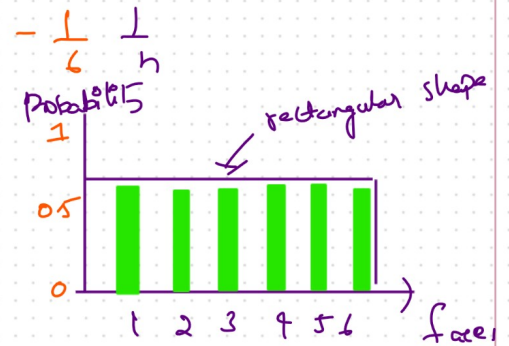


# Statistics

## Uniform distribution: (continuous)

Discrete ← Rolling Dice — (1, 2, 3, 4, 5, 6)



example: random arrival time between 1pm & 2m

1:10 →

1:20 —

—

—

1:10:01

1:20:01

1:10

1:10 — 1:20

pdf



1:10 — 1:10:01

10:00:00

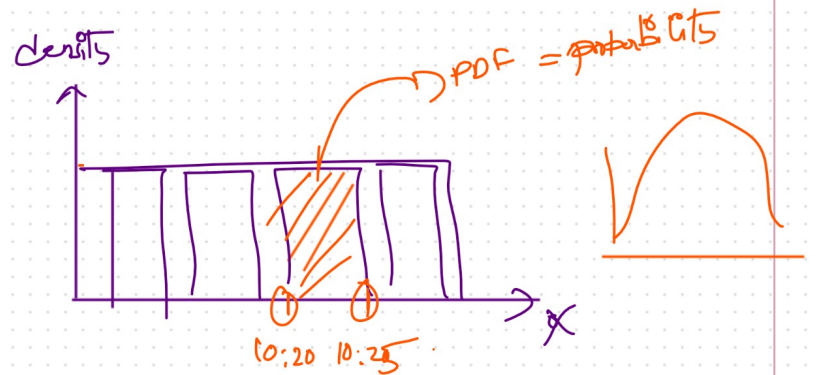
10:00:02

∞ — 10:00 10:05 → minute, m, freq, density

∞ 10:05 10:10

∞ 10:10 10:15

∞ 10:25 10:30

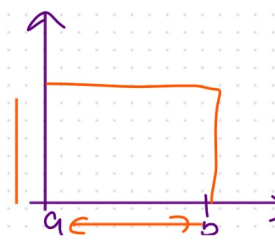


→ Every value in the interval  $[a, b]$  is equally likely

→ The pdf is a flat rectangle, constant height

$$\text{width} = b - a$$

$$\text{height} \times \text{width} = 1 \rightarrow \text{total probability}$$



$$\text{height} = \frac{1}{\text{width}} = \frac{1}{b-a}$$

Scenario A rider can arrive anytime between 20min - 30min

Assume every time in that interval is equally likely

$$\text{So } X \sim \text{Uniform}(20, 30)$$

$a, b$

Question: What is the probability delivery arrives b/w 22 & 25 minutes

$$a=20 \quad b=30$$

$$\text{height} = \frac{1}{b-a} = \frac{1}{10} = 0.1$$

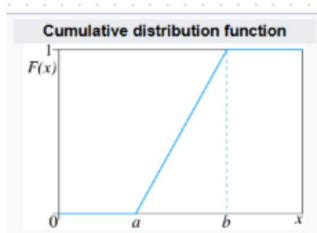
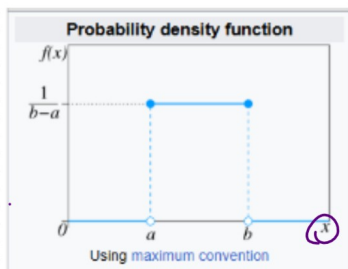
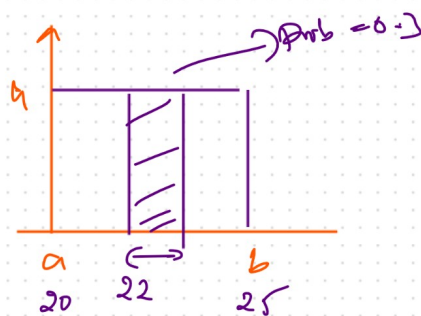
$$P(22 \leq x \leq 25)$$

prob = area of rectangle  
= width  $\times$  height

$$\text{width} = 3 \quad (25 - 22)$$

$$\text{height} = 0.1$$

$$\text{So prob} = \text{width} \times \text{height} \\ = 3 \times 0.1 = 0.3$$



$$\text{cdf} = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in [a, b] \\ 1 & \text{for } x > b \end{cases}$$

Notation  $U(a, b)$

Parameter  $-\infty < a < b < \infty$

$$\text{pdf} = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

$$\text{mean} = \frac{1}{2}(a+b)$$

$$\text{median} = \frac{1}{2}(a+b)$$

$$\text{Variance} = \frac{1}{12}(b-a)^2$$

eg The no. of candies sold daily at a shop is uniformly distributed with a minimum of 40 candies & a maximum of 100.   
 (i) probability of daily sales to fall b/w 15 & 20 (15-30)

## Normal Distribution / Gaussian Distribution

→ The bell-shaped curve

→ Symmetric, Continuous, defined for all real numbers with probability concentrated near the mean

→ It has 2 parameters  $\mu$  (location),  $\sigma$  (spread)

→ In real world normal distribution appears naturally

→ average values happen often

→ extreme values happen rarely

### Example

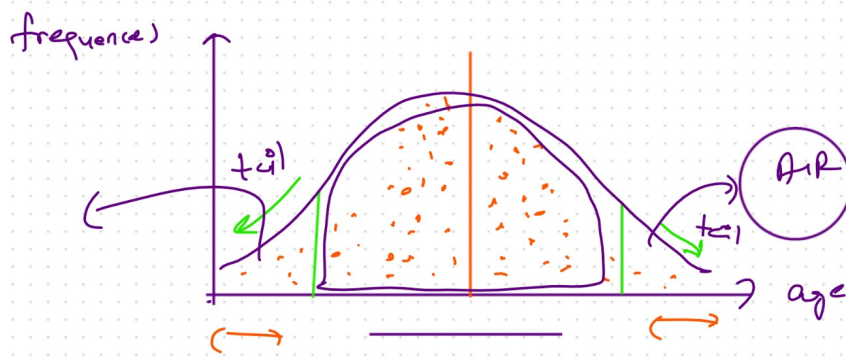
→ human height

→ Exam scores

→ Avg age

Normal Distribution: A model where middle is common & extremes are rare





$$X \sim N(\mu, \sigma^2)$$

Components:

$X$ : variable

$N \rightarrow$  Normal distribution

$\mu$ : mean

$\sigma^2$ : variance

$\sigma$ : STP (spread)

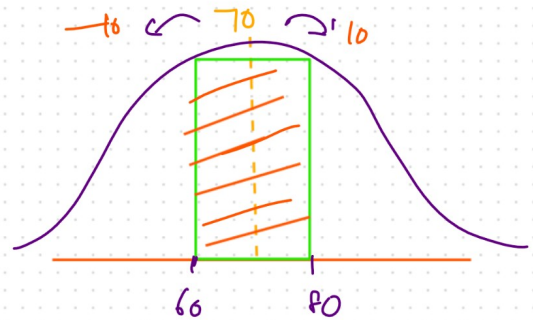
Scenario (fact)

$$X \sim (70, 10^2)$$

$X \rightarrow$  marks in exam

Average marks = 70

Standard deviation = 10



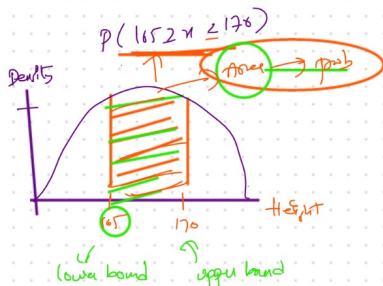
Question: what is the probability a student scores between 60 & 80

PDF  $\rightarrow$  continuous

PMF  $\rightarrow$  discrete

PDF formulae

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$



$$= \int_{60}^{80} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

lowa band  $\rightarrow 60$

upper bound  $\rightarrow 80$

$$z = \frac{x - \mu}{\sigma} = \frac{60 - 70}{10} = -1$$

$$z = \frac{80 - 70}{10} = 1$$

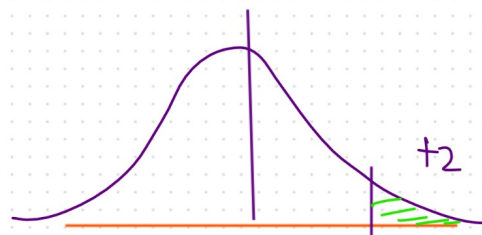
-1 std      +1 std

$$P(-1 \leq z \leq 1)$$

$$P(60 \leq x \leq 80) = 0.6827$$

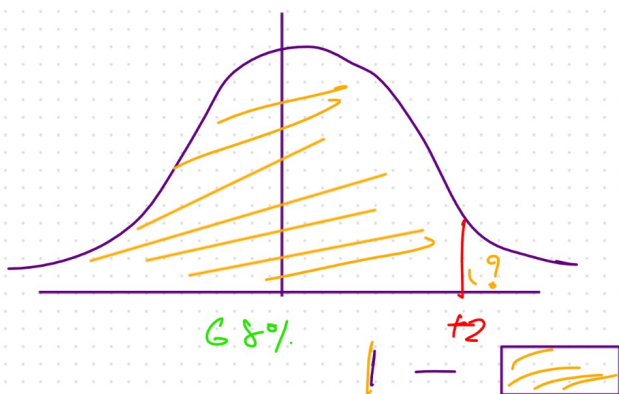
Q) student scores above 90

$$z = \frac{90 - 70}{10} = 2$$



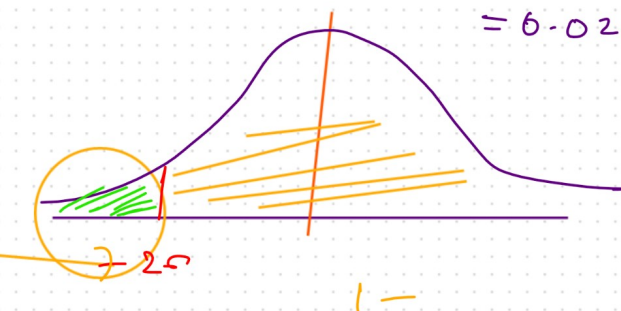
$$\rightarrow P(z \geq 2) = 0.02275$$

$$1 - P(z \geq 2) = 1 - 0.97725 = 0.02275$$



(-68  
 ?

$$z \rightarrow 1.68$$



0.08

1.6

0.95352

probability

$$z \rightarrow -2.94$$

0.04

-2.9

0.00164

prob

# The Empirical Rule (68 - 95 - 99.7) →

within  $\pm 1\sigma$  : 68.27%

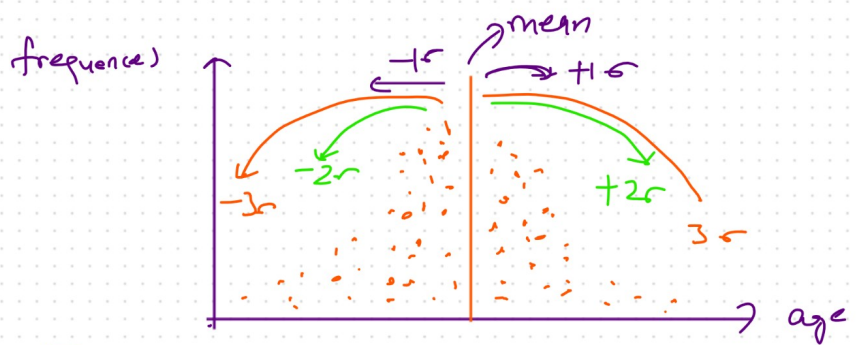
within  $\pm 2\sigma$  : 95.45%

within  $\pm 3\sigma$  : 99.73%

$\pm 4\sigma$  : 99.9937%

$\pm 5\sigma$  : 99.99994%

$\pm 6\sigma$  : 99.999998%



→ much difference

outside of  $\pm 3\sigma$  : 0.27 data extremely rare events

Z-score : How many standard deviations a value is from the mean

$$X = \left\{ \text{---} \text{---} \text{---} \text{---} \text{---} \right\} \leftarrow \text{heights}$$

$$X = \left\{ \text{---} \text{---} \text{---} \text{---} \right\}$$

$x$  → actual value

$\mu$  → mean

$\sigma$  → standard deviation (convert distance into S.D units)

$$Z = \frac{x - \mu}{\sigma}$$

Class A : dist-1 → 100 students → student\_id = 15 → 60 → 65 → 2/100 → 198 → 3.9 → 198 ✓

Class B : dist-2 → 100 students → student\_id → 18 → 60 → 90/100 → top 10 → 80 → last\_rank



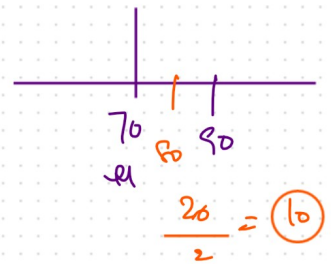
Raw distance alone can mislead

exam A mean = 70, SD = 5

exam B mean = 70, SD = 20

→ from exam A, a student scored 80  
A: 10 (2σ)

B: 10



In A: 2 S.D away

B: 0.5 SD away

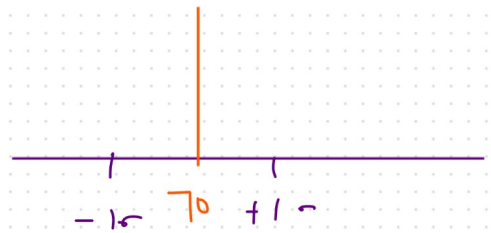
} SD give more content

full example

mean = 70

S.D = 10

Student Scored:  $x = 85$



find z-value:  $\frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = \frac{15}{10} = 1.5$  (above average)

